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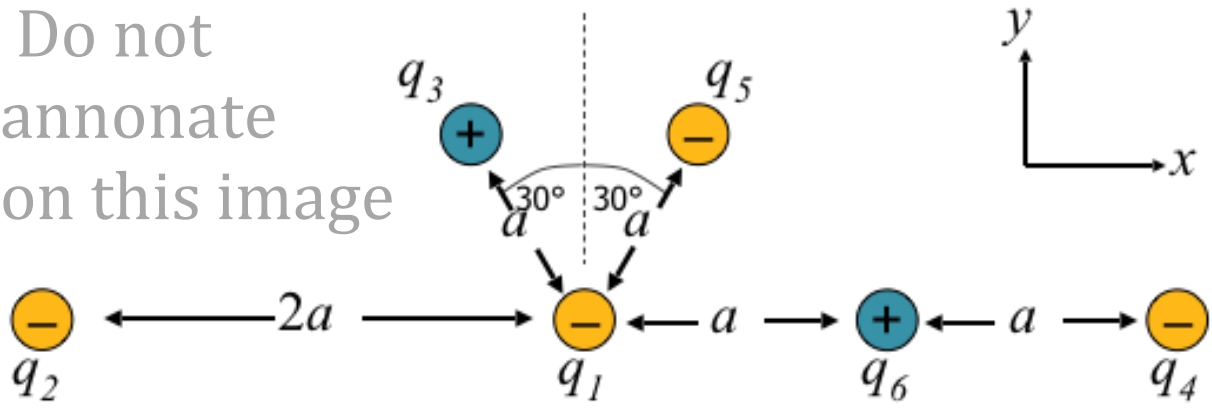
QUESTION 1 Electrostatics - Coulomb **[2 + 2 + 2 + 4 + 4 + (2 + 2) = 18 marks]**

Six point charges are arranged as in the diagram below. All charges are of equal magnitude. Charges q_1, q_2, q_4 and q_5 each have charge $-q$ C, while q_3 and q_6 each have charge $+q$ C. Charges q_1, q_2, q_4 and q_6 are arranged in a straight line at the distances labelled, in metres, whilst q_3 and q_5 are in the same plane as drawn.

Do not annonate on this image



a) Draw and label all the forces acting on q_1, including relative direction and magnitude.

b) Find a functional expression for the net force on q_1 due to the charges in terms of q, a and θ.

c) Evaluate the expression in (b) to find the value of the net force and its direction on q_1.

d) If q_1 is displaced towards the right along the line of charges, give a formula for the net force and direction. [Hint: displace it by δx in the x direction and take derivatives.] Is the charge q_1 trapped?

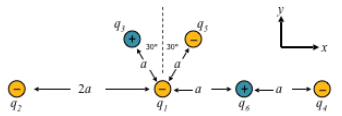
e) Explain why Coulomb's Law obeys $1/r^2$ using a diagram, with reference to concepts of flux and electric field. State the Principle of Superposition and explain why the electrical force obeys the Principle of Superposition. Explain why the electrical force obeys vector addition.

f) i. Draw the electric field lines of an electric dipole.
ii. State the rules you are using in drawing the field lines.

(*Answer Question 1 (a)-(f) in the designated box on the following page.*)

Answers to Question 1(a)-(f)

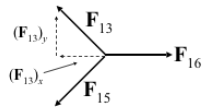
Answers (a) to (c)



Total force on q_1? $\mathbf{F}_1 = \mathbf{F}_{12} + \mathbf{F}_{13} + \mathbf{F}_{14} + \mathbf{F}_{15} + \mathbf{F}_{16}$

Magnitude: $F_{12} = F_{14} = \cancel{k\frac{q^2}{4a^2}}$ But they cancel!

$F_{13} = F_{15} = F_{16} = k_e \frac{q^2}{a^2}$



y components of $\mathbf{F}_{13}$, $\mathbf{F}_{15}$ cancel

$$(F_{13})_x = (F_{15})_x = k_e \frac{q^2}{a^2}\sin 30° = \frac{1}{2}k_e\frac{q^2}{a^2}$$

Hence: $$F_1 = F_{16} - 2\left(\frac{1}{2}k_e\frac{q^2}{a^2}\right) = k_e\frac{q^2}{a^2} - 2\left(\frac{1}{2}k_e\frac{q^2}{a^2}\right) = 0$$

(d)

$$F_x = \overset{12}{k\frac{q^2}{(2a)^2}} - \overset{14}{k\frac{q^2}{(2a)^2}} + \overset{16}{k\frac{q^2}{a^2}} - \overset{13}{k\frac{q^2}{a^2}\underset{1/2}{\sin 30°}} - \overset{15}{k\frac{q^2}{a^2}\underset{1/2}{\sin 30°}}$$
$$= 0$$

$$F_y = \overset{13}{k\frac{q^2}{a^2}\underset{\sqrt{3}/2}{\cos 30°}} - \overset{15}{k\frac{q^2}{a^2}\underset{\frac{\sqrt{3}}{2}}{\cos 30°}}$$

$$F_x' = k\frac{q^2}{(2a+\delta x)^2} - k\frac{q^2}{(2a-\delta x)^2} + k\frac{q^2}{(a-\delta x)^2} - \left\{k\frac{q^2\sin\theta_3'}{\left(a\frac{\sqrt{3}}{2}\right)^2+\left(\frac{a}{2}+\delta x\right)^2}\right\} + k\frac{q^2\sin\theta_5'}{\left(a\frac{\sqrt{3}}{2}\right)^2+\left(\frac{a}{2}-\delta x\right)^2}$$

$$\left(a\cos 30°\right)^3 \quad \left(a\sin 30° + \delta x\right)^3$$

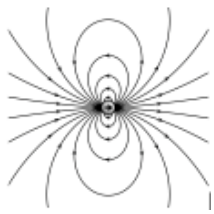
$$F_y' = \{\ \}\cos\theta_3' - \{\ \}\cos\theta_5'$$

$$F_x' = \frac{\partial F}{\partial x}\delta x = kq^2\delta x\left\{\frac{-2}{(2a+\delta x)^3} + \frac{-2}{(2a-\delta x)^3} + \frac{-2}{(a-\delta x)^3} \ldots\right\}$$

x-component – repelled by Hooke's law so acts as a trap in the x-direction.
y-component – if it were three symmetric positive or negative charges it would also trap in the y-direction but in this case q_5 dominates and the force is in the -y direction.

(e) The electric flux through a unit area represents the strength of the electric field and force. The surface area through which the flux travels is 4 pi r^2. Principle of Superposition: forces add, vector fields add, scalar fields add. The electric force obeys the Principle of Superposition because the force is linear with charge. The electrical force obeys vector addition because the functional is linear [Principle of Superposition] and because the field is a vector [so it adds as a vector]

(f) Originate on positive charge [or at infinity].
Terminate on negative charge [or at infinity].
All lines continuous except at source or sink.
Strength of field and force on a charge proportional to the flux [number of field lines] per unit cross-sectional area.



QUESTION 2 Gauss **[(3 + 3 + 3) + (3 + 4 + 2) = 18 marks]**

a) A point positive charge, $+Q$, is placed at the centre of a spherical cavity of radius R formed in a large block of conducting material. Use Gauss' Law to find the magnitude and direction of the electric field at a distance:

i) $r = R/2$, inside the cavity;

ii) $r = R$, at the surface of the cavity;

and iii) $r > R$ from the centre of the cavity, inside the conductor.

b) Two long concentric conducting cylinders of radius a and b, respectively, each carry equal and opposite charges, with the inner shell having a linear charge density $+\lambda$ C m^{-1}. The cylinders are of negligible thickness.
Assuming that $b > a$, use Gauss's law:

i) to show that $\boldsymbol{E}$ vanishes for $r < a$ and $r > b$;

ii) to show that for $a < r < b$, $|\boldsymbol{E}| = \frac{\lambda}{2\pi r \varepsilon_0}$;

and iii) to describe the direction of the electric field in the space between the cylinders.

Answers to Question 2(a)-(b)

$$\int \mathbf{E}.\mathrm{d}A = \frac{q_{\text{enc}}}{\epsilon_0} \tag{31}$$

$$\rightsquigarrow E \times (\text{surface area of sphere}) = \frac{Q_{\text{enc}}}{\epsilon_0} \tag{32}$$

$$\rightsquigarrow E = \frac{Q}{4\pi r^2 \epsilon_0} \tag{33}$$

Substitute $r = R/2$:

$$\rightsquigarrow E = \frac{Q}{\pi R^2 \epsilon_0}. \tag{34}$$

(ii) $r = R$.
Same analysis as above, but substitute $r = R$:

$$E = \frac{Q}{4\pi R^2 \epsilon_0} \tag{35}$$

and the field continues to point **radially outwards**.

(iii) $r > R$ Because the material is conducting, a screening charge forms on the inner surface of the cavity. Thus, any Gaussian surface will have $q_{\text{enc}} = Q + \text{screening charge} = Q + (-Q) = 0$. Thus $\int \mathbf{E}.\mathrm{d}A = 0$. Since we have spherical symmetry, $E = 0$.

b. (i) Consider a cylindrical Gaussian surface of length L and radius r.
For $r < a$, $q_{\text{enc}} = 0$.
For $r > b$, $q_{\text{enc}} = q_a + q_b = L\lambda + (-L\lambda) =$.
Thus:

$$\int \mathbf{E}.\mathrm{d}\mathbf{A} = 0 \tag{36}$$

Symmetry dictates the ends of the cylinder do not contribute.

$$\rightsquigarrow E \times L \times 2\pi r = 0, \qquad L \neq 0, r \neq 0 \tag{37}$$

$$\rightsquigarrow E = 0. \tag{38}$$

(ii) Same shaped Gaussian surface.

$$\int \mathbf{E}.\mathrm{d}A = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0} \tag{39}$$

$$\rightsquigarrow |E| L 2\pi r = \frac{\lambda L}{\epsilon_0} \tag{40}$$

$$\rightsquigarrow |E| = \frac{\lambda}{2\pi r \epsilon_0}. \tag{41}$$

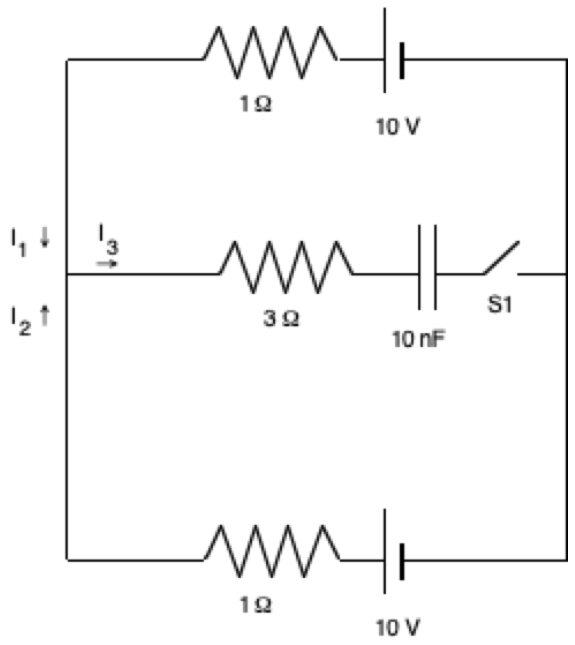
(iii) Consider a cylindrical Gaussian surface, $a < r < b$. As $L \to 0$, $q_{\text{enc}} \to 0$ and so the electric field must have only radial components.

QUESTION 3 Electric Circuits **[(2 + 4) + (1 + 1 + 1) + 1 + (1 + 4) + 1 = 16 marks]**

Use Kirchhoff's rules and symmetry as necessary to consider the following circuit:

a) If switch S1 is open, what is the relation between the current I_1 and I_2 ?

Answer to Question 3(a)



Kirchhoff: junction rule: $I_1+I_2=I_3$. If $I_3 = 0$, then $I_1=-I_2$. From Symmetry $I_1=I_2$, so then $I_1=I_2=0$ A.

The switch S1 is then closed. Immediately after it is closed:

b) i) What is the relation between the current I_1 and I_2? Express I_1 and I_2 in terms of I_3.

ii) What will be the value of current I_3?

iii) What is the voltage drop across the 3 Ω resistor next to the capacitor at this time?

c) After a long time, what charge will have accumulated on the capacitor? What will I_3 be then?

(*Answer Question 1 (a)-(f) in the designated box on the following page.*)

Answers to Question 3(b)-(c)

(b) (i) If switch S1 is closed, Symmetry still valid so $I_1=I_2=I_3/2$.

(ii) C uncharged, so only resistor to discuss. Kirchhoff loop rule: $10V = 1xI_1+3xI_3 = 3.5\ I_3$. $I_3=2.85A$

(iii) Then V=IR so V = 3 x 2.85 = 8.55 V.

(c) I_3 = 0. Fully charged. [1] V=q/C, q=CV=1E-7 C.

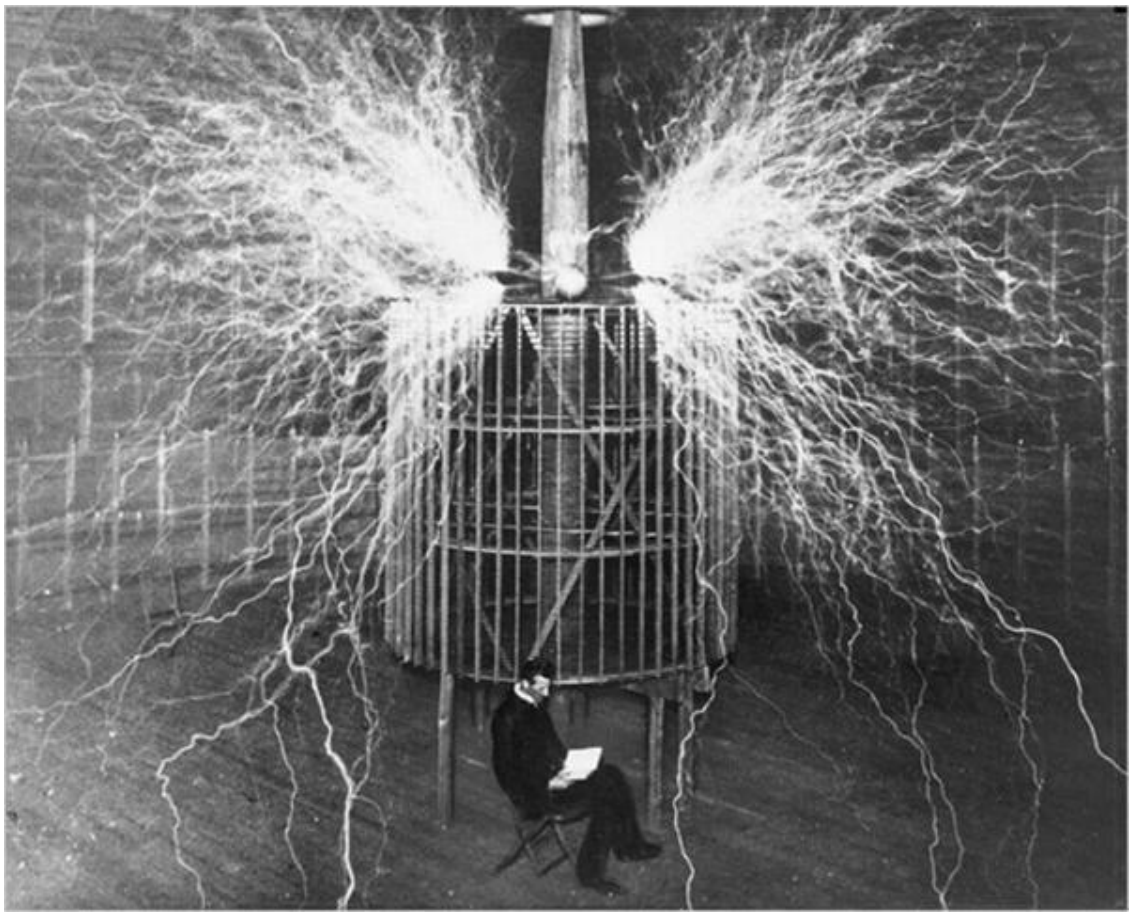
d) Professor Cuthbert Calculus is an absent-minded scientist. You visit him at his laboratory. He invites you to assist him in the following experiments.

i) He has a large conducting rod on a wall, pointing out from the wall, which he has charged up to 10 000 V. You can hear some fizzing and crackling coming from the tip. He switches off the machine and asks you to ground the rod. Assuming that you agree, explain how you would do so safely.

ii) He asks you to keep your hands on the surface of a Van de Graaff accelerator which will be slowly raised to a maximum potential of 100 kV. The sphere on the Van de Graaff is 0.5 m off the floor. The initial current to the sphere is about 100 μA and the sphere accumulates 10 μC at the maximum potential over a period of a few seconds. Air breaks down (ionizes) at $E = 3 \times 10^6$ N/C. Assume any discharge occurs over a period of 1 ms.

The bottom of your shoes has a thickness of 3 mm. Explain what you would expect to occur [Give potential differences and currents in your answer.]

e) Nikola Tesla is pictured below, with discharges of several million volts cascading around him in his Colorado Springs Laboratory. The roar that accompanied such discharges could be heard ten miles away. Tesla is unharmed. Why? Explain in terms of electric field lines and conductors.



(*Answer Question 3 (d)-(e) in the designated box on the following page.*)

Answers to Question 3(b)-(c)

(d) (i) It is safe with a rod of metal to connect from the rod to the ground, if it is insulated from you by e.g. a wooden handle of some useful length. 'Ensure that you do not form part of the circuit.'

(ii) Charge will build up until the potential reaches 3mm x 3M N/C = 9000 V at which it will arc from your foot to ground, in perhaps 0.2ms, giving a current of I = Q/t = 5 or so μC / 1 ms = 5 mA. In this case, you would feel regular arcs, although they would not be dangerous. [it could be worse if it were e.g. 8 mm thick.]

(e) No current passes through him – he does not complete a circuit. He is in a Faraday cage, i.e. he is inside a conducting surface, so no electric fields are inside

QUESTION 4 Ampere **[5 + 2 + 6 = 13 marks]**

a) Using the Biot-Savart Law, show that the magnetic field due to an infinitely long straight wire carrying a current *i* amperes, at a radial distance *r*, has magnitude given by $B = \frac{\mu_0 i}{2\pi r}$ and explain direction.
[Hint: you might find the result $\int \frac{Rds}{[s^2+R^2]^{3/2}} = \frac{s}{R[s^2+R^2]^{1/2}}$ useful.]

b) Show that your answer to part a) above is consistent with Ampere's Law.

Answers to Question 4(a)-(b)

(a) $d\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{i d\boldsymbol{s}\times\boldsymbol{r}}{r^3}$
Integrate Biot-Savart along wire.

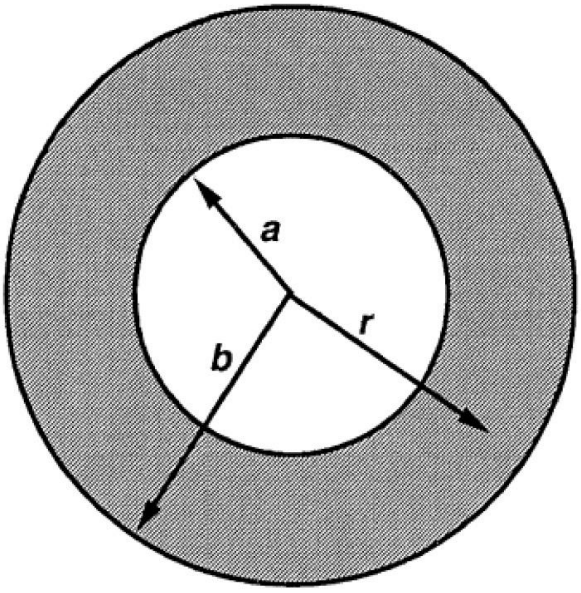
$$\int_{-\infty}^{\infty} \frac{Rds}{[s^2+R^2]^{3/2}} = 2\int_0^{\infty} \frac{Rds}{[s^2+R^2]^{3/2}} = 2\left[\frac{s}{R[s^2+R^2]^{\frac{1}{2}}}\right]_{s=\infty} = \frac{2}{R}$$

So $\int d\boldsymbol{B} = \int \frac{\mu_0}{4\pi}\frac{i d\boldsymbol{s}\times\boldsymbol{r}}{r^3} = \frac{2\mu_0 i}{4\pi r} d\hat{\boldsymbol{s}} \times \hat{\boldsymbol{r}}$

$B = \frac{\mu_0 i}{2\pi r}$ Determines magnitude. Direction = RHR from cross product
(b) $\int \boldsymbol{B}\cdot d\boldsymbol{s} = \mu_0 i = B\, 2\pi r$. loop around wire in circle. Determines magnitude. Direction = RHR

c) Imagine that we have a current-carrying cylinder made from a perfect conductor, with a cylindrical hole through its centre. The inner radius is a m and the outer radius is b m. The current direction is perpendicular to the page in the diagram below. The cross-sectional current density in the cylinder is J Am^{-2}. Indicating any assumptions you make, determine the magnetic flux density [B-field] inside the cylinder for $a<r<b$.

Answer to Question 4 (c)



(c) $i = \pi J(b^2 - a^2)$
inside amperian loop of radius r, $i = \pi J(r^2 - a^2)$

$B(r) = \frac{1}{2}\mu_0 J(r - \frac{a^2}{r})$
Same direction.

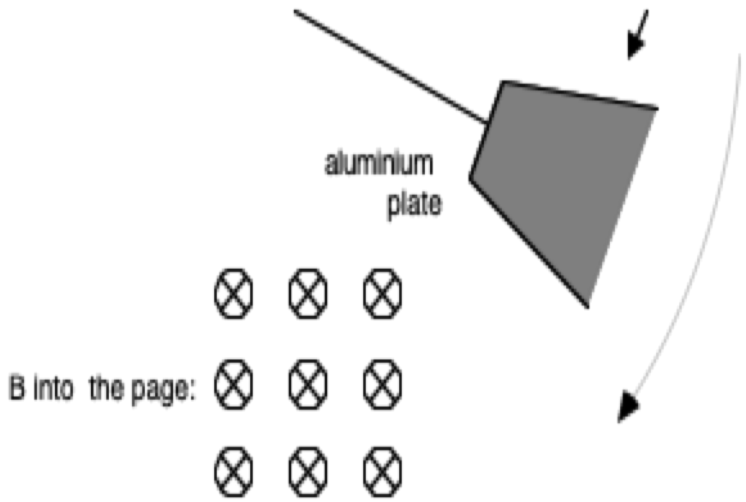
QUESTION 5 Induction **[3 + (3 + 4) = 10 marks]**

(a) A particle of mass m and charge q is moving in a circular path in a magnetic field B. Show that its kinetic energy is proportional to r^2, the square of the radius of its path. [Hint: use the equation for centripetal motion, $F = mv^2/r$]

Answer to Question 5(a)

$F=qvB$, $E=1/2\ m\ v^2$
centripetal motion has $F= mv^2 /r$
so $v = rqB/m$
and $E = 1/2\ [qB]^2/m\ r^2$

(b) (i) A demonstration showed a fan-shaped plate of aluminium swinging through a large 'C' magnet, slowing down on each pass through the magnet and then stopping. Why did the aluminium plate slow down as it entered the magnetic field? Use the RH grip rule in your answer to explain the sign of the effect.



(ii) When a different plate was introduced, with slots in the plate separated by 2 mm, the behaviour of the swinging fan changed dramatically. Compare the number of oscillations of this plate to the behaviour of the plate in (i). Explain in terms of your answer to part (a), eddy currents and the applicable formula describing their cause.

(Answer Question 5 (b) in the designated box on the following page.)

Answers to Question 5 (b)

(i) Magnetic induction creates current loops (eddy currents). [1]

The formula for induced emf in response to changing magnetic flux (Faraday's law) is EMF = - N $\frac{\Delta\Phi_B}{\Delta t}$, $\Phi_B = \Sigma$ B A cos θ. [1]

In the aluminium tube, T>>the time to fall under gravity

By Lenz's law these oppose the change in magnetic flux as the magnet falls through. Hence the induced fields oppose the motion and slow the magnet down dramatically.

(ii) The formula for induced emf in response to changing magnetic flux (Faraday's law) is EMF = - N $\frac{\Delta\Phi_B}{\Delta t}$, $\Phi_B = \Sigma$ B A cos θ. In this case the field is varying on any region of the plate surface. Here N=1. [same as i].

Because it is a conductor the EMF drives a circuit EMF = IR where R is some effective resistance of the aluminium. Hence a current loop forms and following the RH Grip rule produces a field opposing the original change of flux $\sum_{loop} B.\Delta l = \mu I$.

By Lenz's law these oppose the change in magnetic flux as the magnet falls through. Hence the induced fields oppose the motion and slow the magnet down dramatically. F = IlB sin theta (for the side l in the field and perpendicular to the motion of the plate).

The motion is therefore damped by induction (eddy currents) induced in the plate.

The plate's speed is much less damped. The emf induced is reduced, because the radius of the loops (i.e. the maximum area A) is restricted.

The cuts in the plate are open circuits for any large current loops that might otherwise be formed. Now the electrons cannot go around in a large circle in the plane of the plate (i.e. perpendicular to the field change)

QUESTION 6 Fluids **[2 + 3 + 3 + 3 = 11 marks]**

An artificial lake is contained by a dike. This dike has developed a circular hole of 1.2 cm diameter. If hole is 3 m below the water level

i) What is the force exerted on this hole by the water?

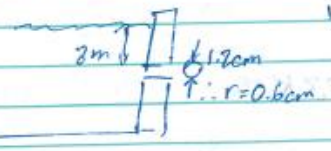
ii) How long would take for the released water to fill 2 acres of land (2,468 m^2) to a depth of 1 m? Assume the density of the water is 1.0 kg/m^3 and that the hole remains constant in size.

iii) What will be the pressure exerted on the flooded area by the water?

iv) In a flood, objects may be displaced. A big box with a mass of 500 kg and a volume of 1 m^3 lies at the bottom of the flooded area. How much force is needed to lift it at *constant velocity*?

Answers to Question 6 (i) – (iv)

6 Fluids



i) $P = P_{atm} + \rho g h$

$= 101325 + 1000 \times 9.8 \times 3$

$= 130725\ Pa$ (or just 29400 Pa if by water alone)

$(V = 2468 \times 1 = 2468\ m^3)$

$P = \frac{F}{A} \Rightarrow 130725 \times (\pi \times (0.6 \times 10^{-2})^2) = F = 14.78\ N$

(or $F_{water} = 3.33\ N$)

ii)

$\rightarrow P_{atm} + \rho g h = \frac{1}{2}\rho v^2 + P_{atm}$

$1000 \times 3 \times 9.8 = \frac{1}{2} \times 1000 \times v^2$

$\therefore v = 7.67\ ms^{-1}$

$\rightarrow$ Volume/sec $= Av = \pi(0.6 \times 10^{-2})^2 \times 7.67$

$= 8.67 \times 10^{-4}\ m^3/s$

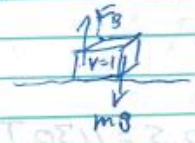
$\therefore \frac{V}{Q} = \frac{2468}{8.67 \times 10^{-4}} = 790\frac{1}{2}$ hours $= 2845798\ s$

$= 32.9$ days

iii) $P_{flooded} = 1 \times 9.8 \times 1000$

$= 9800\ Pa$

iv)



$F_B = \rho_{water} V_{obj} g$

$= (1000\ kg/m^3)(1\ m^3)(9.8) = 9800\ N$

$mg = 500 \times 9.8 = 4900\ N$

need OPPOSING Force of 4900 N DOWN to stop box accelerating as it floats to surface.

Answers to Question 6 (i) – (iv) - continued

QUESTION 7 Heat and Work **[(2+ 2 + 2) + 2 + 3= 11 marks]**

a) A cylinder of Nitrogen (diatomic molecule) gas has a volume of 15,000 cm^3 and a pressure of 100 atm.

b)

i) What is the thermal energy of this gas at room temperature (20 °C)?

ii) What is the mean free path in the gas ($r = 1.0 \times 10^{-10}$ m) ?

iii) The valve of the cylinder is open and the gas is allowed to expand slowly and isothermally until it reaches 1 atm. What is the change in the thermal energy of the gas?

b) Even if a person shows no visible perspiration, this person will still evaporate about 500 g of water from their lungs daily. How many Joules of heat are removed by this evaporation? What is the rate of heat loss in watts due to this process?

heat of vaporization of water is 2260 kJ/kg

c) In the kinetic theory of gasses explain what is meant by the "equipartition of energy".

Answers to Question 7 (a)-(c)

7.

$V = 15,000\ cm^3 = 0.015\ m^3$

$P = 100\ atm = 10,132,500\ Pa$

$T = 20°C = 293\ K$

Diatomic Nitrogen

a)

i) Thermal Energy = Kinetic Energy? $= \frac{5}{2} k_B T N$

$PV = Nk_BT \rightarrow \frac{10132500 \times 0.015}{1.38 \times 10^{-23} \times 293} = 3.76 \times 10^{25} = N$

$\therefore E = N\bar{K} = 3.8 \times 10^5\ J$

ii) $\lambda = \frac{1}{4\sqrt{2}\pi r^2 \eta}$, $\eta = \frac{N}{V}$

$= \frac{1 \times 0.015\ m^3}{4\sqrt{2}\pi(10^{-10})^2 \times 3.76 \times 10^{25}} = 2.25 \times 10^{-9}\ m$

iii) $PV = nRT = Nk_BT$

Isothermally $\therefore \Delta T = 0 \therefore \Delta E = 0 \therefore \Delta Q = \Delta W$

$W = -Nk_BT \log_e \frac{V_2}{V_1}$

$P_1V_1 = P_2V_2 \therefore 100 \times 0.015 = 1 \times V_2 = 1.5\ m^3$

$\therefore W = -Nk_BT \log_e(100) \approx -700\ 132\ J$

$\therefore Q = +700\ 132\ J$

b) $m = 500\,g$ water $\quad Q = mL = 2260\,kJ \times 0.5 = 1130\,J$

$P = \frac{dQ}{dt}$ but $\Delta t = ?$

c) Equip^{n} Theory - each degree of freedom in a gas carries $+\frac{1}{2}k_BT$ worth of thermal energy

i.e. monatomic gas has $3/2\,k_BT$, diatomic $5/2\,k_BT$

(moves x, y, z) (moves x, y, z, vibrat^{n}, rotat^{n})

QUESTION 8 Particle wave duality **[(2 + 2 + 3) + (2 + 2) + 3 + (2 + 2) = 18 marks]**

a) The work function for rubidium is 208.4 kJ/mol.

i) What is the minimum amount of energy needed to eject a single electron from the surface of rubidium?

ii) What is the maximum wavelength, in nm, that can produce photoelectric current from rubidium?

iii) If a beam of green light with wavelength 550 nm strikes the surface of rubidium, will electrons be ejected? If yes, calculate the kinetic energy and speed of the ejected electron.

b) In a Compton collision with an electron inside rubidium, a photon of violet light (λ = 4000Å) is backscattered through an angle of 180°.

i) How much energy (in eV) is transferred to the electron in this collision?

ii) Could violet light eject electrons from the rubidium? Explain.

c) An electron and a proton are accelerated to 25.0% the speed of light. What are the wavelengths associated with each particle in nm? (m_e = 9.11 x 10^{-31}kg; m_p = 1.67 x 10^{-27} kg;)

d) An electron moves with a speed of 1.5 x 10^7 m/s, with uncertainty on its speed of 1 %.

i) What is the absolute value on the uncertainty of the electron's speed? If the location of the electron is simultaneously measured, what is the minimum uncertainty (Δx) in knowing the electron's location?

ii) Electrons have a diameter of about 10^{-17} m. Compare this with Δx, what can you conclude?

Answers to Question 8 (a)-(d)

8. Particles

a) i) $E_0 = 208.4\ kJ/mol$

$\therefore KE_{min} = hf - E_0 = 0$

$\therefore hf = E_0 = 208.4\ kJ/mol = 208000/6.022\times10^{+23}$
$= 3.46\times10^{-19}\ J$ per atom ($\because$ per e^-?)

ii) λ_{max}? $E_{min} = E_0$
$= \frac{hc}{\lambda_{max}} \Rightarrow = \frac{hc}{E} = 574.7nm$

iii) If $\lambda = 550nm$, yes to P.E. Effect

$\rightarrow KE = \frac{hc}{550nm} - (3.46\times10^{-19}\ J)$
$= 1.56\times10^{-20}\ J$
$= 0.098\ eV$

b) i) $\gamma = 400nm$, $\theta = 180°$, e^-

$\Delta\lambda = \frac{h}{m_e}(1-\cos\theta) = \frac{2h}{m_e c}$
$= 4.85\times10^{-12}\ m$

$\therefore \lambda_{new} \approx 400\ nm$

$\therefore p_\gamma = \frac{h}{\lambda} = 1.6575\times10^{-27}\ kg\,ms^{-1}$

& $p_\gamma' = -p_\gamma = -1.6575\times10^{-27}\ kg\,ms^{-1}$

$\therefore e^-$ gains $2p_\gamma (= \Delta p) = 3.31\times10^{-27}\ kg\,ms^{-1}$

$\rightarrow E_e = \frac{p^2}{2m_e} = 6.03\times10^{-24}\ J$

c) $v_p = v_e = 0.25c \rightarrow \lambda_e, \lambda_p$? $\quad p = \frac{h}{\lambda} = \gamma m v$

At $v = 0.25c$, $\gamma = 1.033$, $p_e = 7.06 \times 10^{-23}\ kgms^{-1}$
$p_p = 1.29 \times 10^{-19}\ kgms^{-1}$

$\therefore \lambda_e = \frac{h}{p_e} = 9.39\ pm = 9.39 \times 10^{-3}\ nm$

$\lambda_p = \frac{h}{p_p} = 5.14\ fm = 5.14 \times 10^{-6}\ nm$

Q8 (Continued)

d) i) $v = 1.5 \times 10^{7}\ m/s \rightarrow \Delta v = \underline{1.5 \times 10^{+5}\ m/s}$

$p = mv = 9.11 \times 10^{-31} \times 1.5 \times 10^{7}\ ms^{-1}$
$= 1.37 \times 10^{-23}\ kgms^{-1}$

$\frac{\Delta v}{v} = 0.01 = \frac{\Delta p}{p} \quad \therefore \Delta p = 1.37 \times 10^{-25}\ kgms^{-1}$

$\rightarrow \Delta x \Delta p \geq \hbar/2$

$\Delta x \geq \frac{\hbar}{2\Delta p} = \underline{3.86 \times 10^{-10}\ m}$

ii) Can't know e^- any smaller than 10^{-10} m, to determine if really 10^{-17} m → would need more inaccurate/uncertain momentum value Δp for better Δx.

QUESTION 9 The black body **[2 + (2 + 2 + 2) = 8 marks]**

a) The peak in the frequency spectrum of the so-called Cosmic Microwave Background (CMB) radiation, the relic radiation from the early stage of the universe, is f_{peak} = 160 GHz. What is the temperature of the CMB?

b) Venus' surface temperature is about 750 °C

   i) Estimate λ_{max} of the emitted radiation by Venus.

   ii) Venus is a sphere of 6×10^3 km radius, how much power does it radiate?

   iii) Find the spectral power per wavelength at λ_{max}.

Answers to Question 9 (a)-(b)

Q9 a) CMB $f_{peak} = 160\times10^9 Hz$

$\rightarrow c = \lambda f \rightarrow \lambda = \frac{c}{f} = 0.001875 m$

$T = \frac{2.898\times10^{-6}}{\lambda} = \underline{\underline{1.5456 K}}$

b) i) $T_{venus} \approx 750°C$ (!! → not true, ~470°C !!)

$\lambda_{max} = \frac{2898\times10^{-6}}{1023.15K} = \underline{\underline{2.83 \mu m}}$

ii) $r_f = 6\times10^3 km \rightarrow SA = 4\pi r^2 = 4.52\times10^{14} m^2$

$\varepsilon = 1 \quad \sigma = 5.67\times10^{-8} Wm^{-2}K^{-4}$

$\rightarrow P = \sigma\varepsilon A T^4$

$= 2.81\times10^{19} W$

iii) $S_\lambda(750°C) = \frac{2\pi c^2 h}{\lambda^5}\left(\frac{1}{e^{\frac{hc}{k_B T\lambda}}-1}\right)$

$= 1.44\times10^{10}$ flux/λ_{max}

Answers to Question 9 (a) – (b) - continued

QUESTION 10 The atom **[3+ 1 + 2 + (2 + 2 + 2+ 3) = 15 marks]**

a) In classical physics an atom would not be stable because it would emit electromagnetic waves of frequency *f*, because the electron's orbit, seen edge on, can be seen as an electric dipole. At what electron radius, in nm, would the electron emit light with a wavelength of 600 nm?

b) Why in an atom are the energies of the electrons negative?

c) A hydrogen atom makes a transition from the n = 2 to the n = 1 energy level. Find the wavelength and frequency of the emitted photon.

d) In interstellar space, highly excited hydrogen atoms, called Rydberg atoms, have been observed. These can be described by standard Bohr atom theory.
i) Find the wavelength radio astronomers must tune to, so they can detect signals from electrons dropping from the n = 273 level to the n = 270 level.
ii) What is the radius of the Rydberg atom for which n = 273?
iii) What is the speed of an electron in this Rydberg atom?
iv) Compare the results in ii) and iii) with the radius and velocity of hydrogen atoms in their lowest state. What did you learn from this comparison?

Answers to Question 10 (a)-(d)

Q10 : The Atom

a) $\lambda_{em} = 600\,nm \rightarrow E = \frac{hc}{\lambda} = 3.3 \times 10^{-19}\,J$

$E = K + U = \frac{1}{2}mv^2 - k\frac{q^2}{r}$

$\frac{E}{r} = \frac{1}{2}\left(\frac{mv^2}{r}\right) - k\frac{q^2}{r^2}$

$r_{orbit} = ?$ such that $\lambda_g = 600 \times 10^{-9}\,m$

$\frac{E}{r} = \frac{1}{2}\left(k\frac{qq}{r^2}\right) - k\frac{q^2}{r^2} = -\frac{3}{2}k\frac{q^2}{r^2}$

$E = -\frac{3}{2}k\frac{q^2}{r}$

$\rightarrow r = \left|-\frac{3}{2}k\frac{q^2}{E}\right| = \underline{1.05 \times 10^{-9}\,m}$

b) Energies negative as these are binding Energies that need to be overcome to release an electron — when an electron is free it has 0 potential energy, ∴ in the atom, it must have −ve potential E.

c) $n_1 = 2 \rightarrow n_2 = 1$

$$f = cR\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right) = 3\times10^8 \times 1.07\times10^7\left(1 - \frac{1}{4}\right)$$
$$= \underline{2.47\times10^{15}\ Hz}$$
$$\lambda = \frac{c}{f} = \underline{121.5\ nm}$$

d) i) $f = cR\left(\frac{1}{(270)^2} - \frac{1}{(273)^2}\right) = 9.86\times10^8\ Hz = \underline{986\ MHz}$

$\rightarrow \lambda = 0.304\ m = \underline{30.4\ cm}$

ii) $r_{273} = n^2 a_0 = (273)^2 \times 52.92\times10^{-12}\ m = \underline{\underline{3.94\ \mu m}}$!!

iii) $v_{273} = c\alpha_{273} = \frac{3\times10^8}{137\times273} = \underline{\underline{8015\ ms^{-1}}}$

iv) $v_H = 2.2\ Mms^{-1}$ $r_H = 52.92\ pm$ → Rydberg atoms <u>much</u> <u>larger</u>, but then <u>much</u> <u>slower</u>.

QUESTION 11 The nucleus, nuclear decay [(1 + 2) + (3 + 4) + 2 = 12 marks]

The nuclear reactor at ANSTO creates the isotope Mo-99 which then decays to the metastable isotope Tc-99m. The Tc-99m is used as a radioactive tracer for helping to determine the cause of a patient's illness. The Tc-99m tracer emits a gamma ray during its decay to Tc-99. Tc-99 then further decays to Ru-99 which is a stable isotope.
The decay series and relevant information for each isotope is given in the table below;

Name		**Isotope**	**Mass (u)**	**Half-life**	**Decay**
Molybdenum	Mo-99	${}^{99}_{42}Mo$	98.9077119	2.7474 days	Beta
Metastable Technetium	Tc-99m	${}^{99m}_{43}Tc$	98.9066984	6.0058 hrs	Gamma
Technetium	Tc-99	${}^{99}_{43}Tc$	98.9062547	211,000 yrs	Beta
Ruthenium	Ru-99	${}^{99}_{44}Ru$	98.9059393	stable	

Rest masses of: ${}^{4}_{2}He$ atom: 4.002602 u; electron: 0.000549 u; positron: 0.000549 u
${}^{1}_{1}H$ atom: 1.007825 u; neutron: 1.008665 u; proton: 1.007276 u

(a) The Molybdenum is transported to hospitals and produces the metastable Technetium to be injected into the patient.

(i) Mo-99 decays to the metastable Tc-99m via beta decay. Write the decay equation and justify whether the decay is β- or β+.

(ii) Show that the energy released during this decay is 1.216 MeV.

(b) To diagnose a patient, they are injected with the metastable Technetium which has been separated from the Molybdenum. The resulting gamma rays emitted as the isotope decays is an excellent radioactive source for imaging.

(i) Calculate the percentage of Tc-99m that will still be present in the patient's body after 1 day.

(ii) Both Mo-99 and Tc-99 are beta sources, yet only Tc-99 remains in the patient's body as Mo-99 is removed from the sample before the procedure. Using the data in the table, compare the amount of radiation received from each of the sources and comment on why it is fine for Tc-99 to remain in the body, but not Mo-99. Include calculations to justify your answer.

(c) Ruthenium is the only stable atom in the decay series. Determine the binding energy per nucleon of Ru-99.

Answers to Question 10 (a)-(d)

Refer to Q14 from Week 12: PSC 11 (see next page for tutorial answer)

14 (a)i. ${}^{99}_{42}Mo \rightarrow {}^{99m}_{43}Tc + {}^{0}_{-1}\beta + \bar{\nu}$ Beta-minus decay illustrated by A & Z numbers.

14 (a)ii. (Atomic mass of Mo) = 98.9077119u (Atomic Mass of TcM) = 98.9064065u

$$\rightarrow m_{Mo} = 98.9077119\text{u} \times 931.49\frac{\text{MeV}}{\text{u}} = 92.131\text{ GeV}$$

$$m_{Tc(m)} = 98.9064065\text{u} \times 931.49\frac{\text{MeV}}{\text{u}} = 92.1303286\text{ GeV}$$

$$\Delta E = (92.131545 - 92.1303286) \times 10^3\text{ MeV} = \mathbf{1.216\ MeV}$$

Note: Atomic masses *include* electron masses, so the mass of the beta particle is included within the value of the Tc(m) and does not need to be counted again.

14 (b) BE/c² = $\left[\left(44m_p + 55m_n + 44m_e\right) - m_{Ru}\right]$

$$= [(1.65777 \times 10^{-25}kg) - (98.905393 \times 1.6605 \times 10^{-27}kg)]$$

$$\rightarrow \text{BE} = 1.54 \times 10^{-27} \times c^2 = 1.3584 \times 10^{-10}\text{J} = 865.1\text{ MeV}$$

$$\therefore \frac{\text{BE}}{A} = \frac{865.1\text{ MeV}}{99} = \mathbf{8.738\ MeV\ per\ nucleon}$$

Formula Sheet and Useful Data for PHYC10004 (may be detached)

Constants

Mass of electron, m_e	9.109×10^{-31} kg = 0.00055 *u*	Atomic mass unit *u*	1.6605×10^{-27} kg = 931.49 MeV/c^2
Mass of proton, m_p	1.673×10^{-27} kg = 1.00728 *u*	1 eV = 1.6×10^{-19} J	
Mass of neutron, m_n	1.675×10^{-27} kg = 1.00866 *u*	Speed of light in vacuum, *c*	299 792 458 m s^{-1}
Electron charge magnitude, *e*	1.6×10^{-19} C	Gravitational acceleration, *g*	9.8 m s^{-2}
Permeability of free space, μ_o	$4\pi \times 10^{-7}$ T m A^{-1}		
Permittivity of free space, ε_o	8.85×10^{-12} C^2 N^{-1} m^{-2}	Planck's constant, *h* = 6.63×10^{-34} J s = 4.14×10^{-15} eV s	
Bohr radius, a_B	0.529×10^{-10} m	Rydberg constant, *R*	1.097373×10^{7} m^{-1}
Universal gas constant, *R*	8.31 J K^{-1} mol^{-1}	Boltzmann constant, *k*	1.38×10^{-23} J K^{-1}
Avogadro's number, N_A	6.02×10^{23} mol^{-1}	Stefan-Boltzmann constant, σ	5.67×10^{-8} W m^{-2} K^{-4}
Specific heat capacity of ice	2220 J K^{-1} kg^{-1}	1 atm = 101325 Pa = 760 mm Hg	
Specific heat capacity of water	4187 J K^{-1} kg^{-1}	Latent heat of fusion for water	333 kJ kg^{-1}

Formulae

Electromagnetism		
$\boldsymbol{F} = \frac{q_1 q_2}{4\pi\varepsilon_o r^2}\hat{r}$	$\boldsymbol{E} = \frac{\boldsymbol{F}}{q}$	$V(P) = \int_P^{\infty} \boldsymbol{E}.d\boldsymbol{s} = \frac{U(P)}{q}$
$W = -\Delta U$	$\boldsymbol{E} = \frac{q}{4\pi\varepsilon_o r^2}\hat{\boldsymbol{r}}$	$V = \frac{q}{4\pi\varepsilon_o r}$
$\oint \boldsymbol{E}.d\boldsymbol{A} = \frac{q_{enc}}{\varepsilon_o}$	$\Delta V = -\int_{P_2}^{P_1} \boldsymbol{E}.d\boldsymbol{s}$	$\boldsymbol{p} = q\boldsymbol{d}$ $\boldsymbol{\tau} = \boldsymbol{p} \times \boldsymbol{E}$ $U = -\boldsymbol{p}.\boldsymbol{E}$
$V = iR$	$P = iV = \frac{V^2}{R} = i^2 R$	$\frac{V_S}{V_p} = \frac{N_S}{N_p}$
$C = \frac{Q}{V}$	$U = \frac{1}{2}qV = \frac{1}{2}CV^2 = \frac{1}{2}\frac{q^2}{C}$	$C = \frac{A\varepsilon_o}{d}$
$\varepsilon = \kappa\varepsilon_0$	$U_E = \frac{1}{2}\kappa\varepsilon_o E^2$	$i = \frac{dq}{dt} = enAv_d$
$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$	$\boldsymbol{F} = i\boldsymbol{l} \times \boldsymbol{B}$	$\oint \boldsymbol{B}.d\boldsymbol{A} = 0$
$B = \mu_o n i$	$\Phi_E = \int \boldsymbol{E}.d\boldsymbol{A}$	$\Phi_B = \int \boldsymbol{B}.d\boldsymbol{A}$
$d\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{i d\boldsymbol{s} \times \hat{\boldsymbol{r}}}{r^2}$	$\varepsilon = \oint \boldsymbol{E}.d\boldsymbol{s} = -\frac{d\Phi_B}{dt}$	$\oint \boldsymbol{B}.d\boldsymbol{s} = \mu_o i_{enc} + \mu_o i_D$
$i_D = \varepsilon_o \frac{d\Phi_E}{dt}$	$\mu = i \times (area\ of\ loop)$	$\boldsymbol{\tau} = \boldsymbol{\mu} \times \boldsymbol{B}$ $U = -\boldsymbol{\mu}.\boldsymbol{B}$

Fluids and Thermal Energy		
$R = A_1 v_1 = A_2 v_2$	$p + \rho g h + \frac{1}{2}\rho v^2 = const$ $p = p_{Atm} + \rho g h$	$\lambda = \frac{1}{4\sqrt{2}\pi\,(N/V) r^2}$
$pV = nRT$	$\rho = \frac{M}{V}$	$Q = mL$ $Q = mc\Delta T$
$pV^\gamma = const$ $\gamma = \frac{c_p}{c_V} \quad c_p = \frac{f}{2}R$	$U = \frac{f}{2}nRT$ $Q = nc_p\Delta T$	$\Delta U = Q + W$
$W = -\int p dV = -p\Delta V$	$W = -nRT\ln\frac{V_f}{V_i}$	$v_{RMS} = \sqrt{\frac{3kT}{m}}$

Modern Physics		
$\lambda_{MAX} T = 2898\,\mu m\,K$	$u_\lambda(T) = \frac{8\pi c h}{\lambda^5}\left(\frac{1}{e^{\frac{hc}{kT\lambda}} - 1}\right)$	$s_\lambda(T) = \frac{c}{4}u_\lambda(T)$
$P = \sigma\varepsilon A T^4$	$hf = K_{MAX} + \phi$	$E = hf = \frac{hc}{\lambda}$
$\lambda = \frac{h}{p}$	$c = f\lambda$	$\Delta\lambda = \frac{h}{mc}(1 - \cos\theta)$
$E_n = \frac{h^2}{8mL^2}n^2$	$r_n = \frac{4\pi\epsilon_o\hbar^2}{me^2}n^2 = a_o n^2$	$\Delta x \Delta p \geq \frac{\hbar}{2}$
$E_n = -\frac{me^4Z^2}{8\varepsilon_0^2 h^2 n^2} = -\frac{13.6Z^2}{n^2}$	$a_o = \frac{\epsilon_o h^2}{\pi m e^2} = 52.92\,pm$	$v_n = c\alpha_{em}\frac{1}{n}$
$f = cR\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$	$r = (1.2 \times 10^{-15} m)A^{\frac{1}{3}}$	$\alpha_{em} = \frac{e^2}{4\pi\epsilon_o\hbar c} \approx \frac{1}{137}$
$N(t) = N_0 e^{-\lambda t}$	$\lambda = \frac{\log_e 2}{\tau_{1/2}}$	$\frac{B.E.}{c^2} = (A - Z)m_n + Zm_p - m_{nucleus}$